

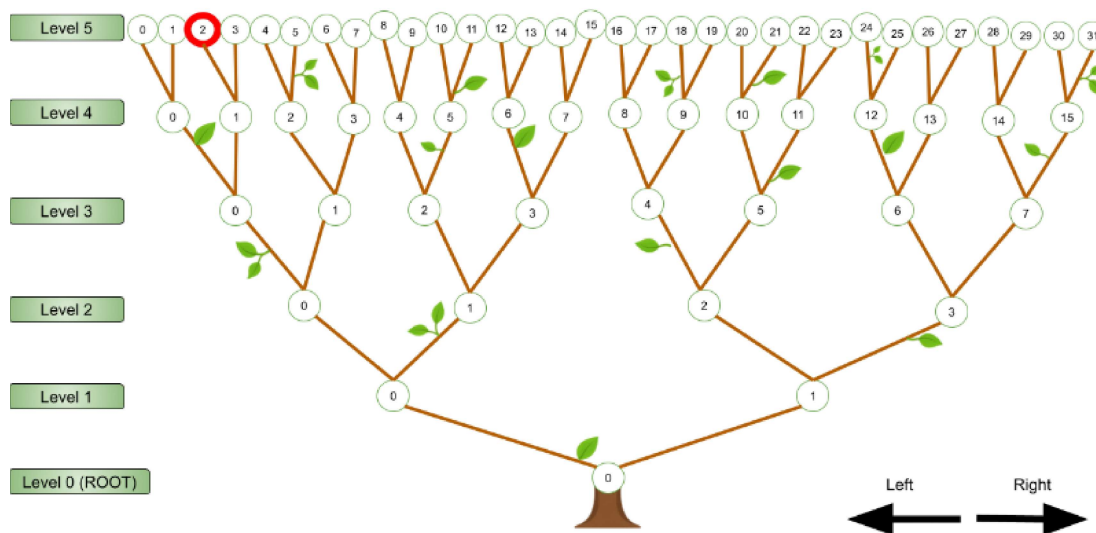
Section 1

Fruit-tree is an environment where the goal is to reach one of the ends of a binary tree starting from the root to pick up a fruit that has multiple different nutritional values (see image, the **RED** color represents where the agent is. The agent in the image is on LEVEL 5, POSITION 2).

The agent always starts on the root and needs to decide if it goes down left or right. Once the agent reaches the end, it receives the reward.

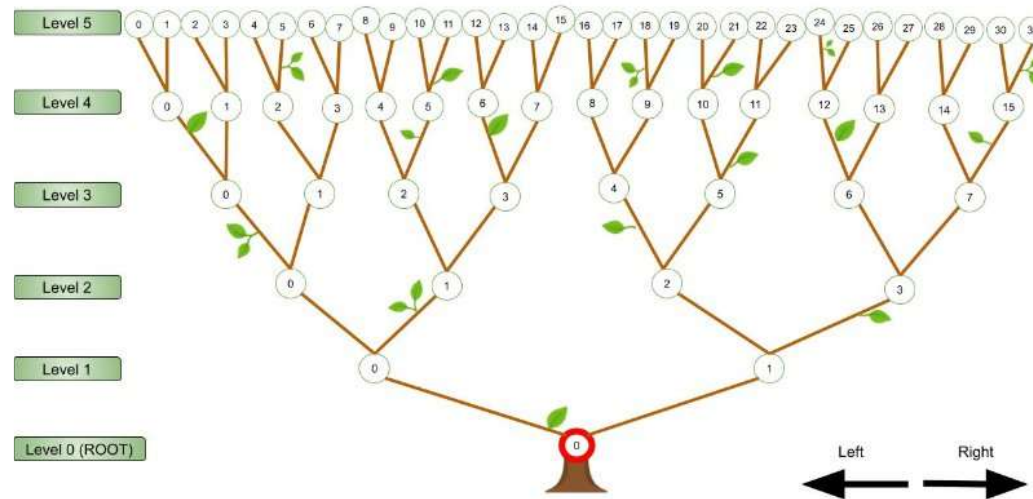
To decide on which path to take, an agent has a static preference for different nutritional values to different degrees. The agent can also observe at which vertical level of the tree it currently is, and which horizontal position it currently is at.

The root of the tree is level 0 and the end of the tree is level 5. At each level, nodes are assigned horizontal positions starting from 0 at the leftmost node and increasing by 1 for each node to its right.



11

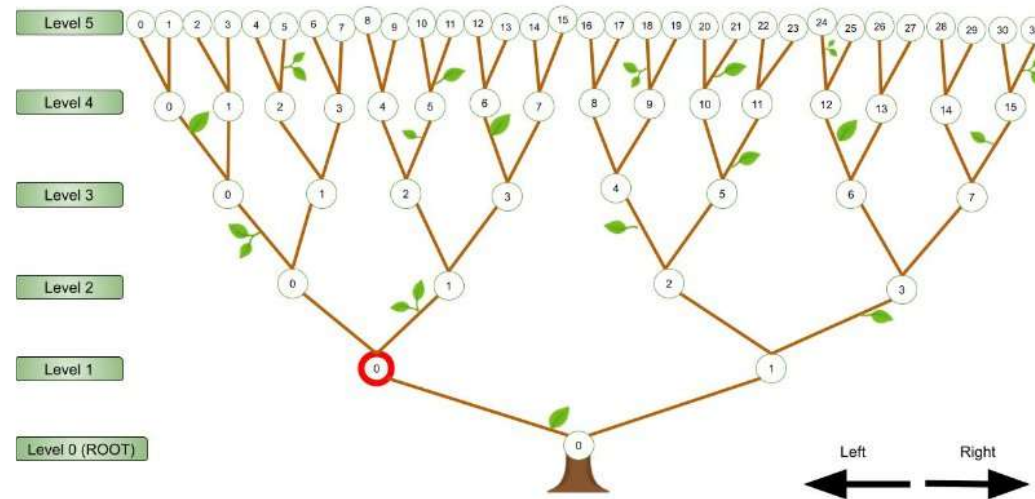
In the tree shown **BELLOW**, at which **LEVEL** is the agent currently located? *



- ☐ Level 0
- ☐ Level 1
- ☐ Level 2
- ☐ None of the above

12

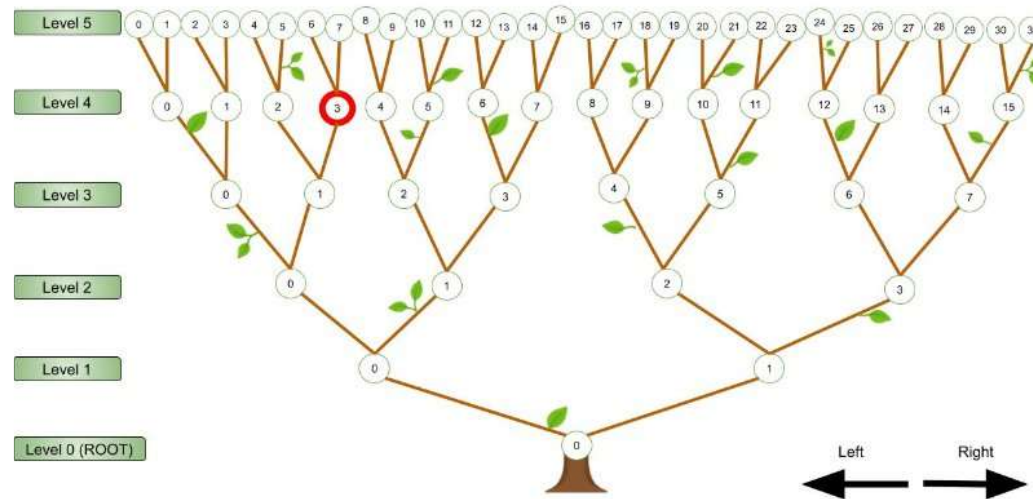
In the tree shown **BELLOW**, at which **POSITION** is the agent currently located? *



- ☐ Position 0
- ☐ Position 1
- ☐ Position 2
- ☐ None of the above

13

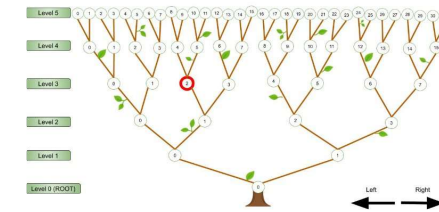
In the tree shown **BELLOW**, at which **POSITION** is the agent currently located? *



- ☐ Position 0
- ☐ Position 3
- ☐ Position 5
- ☐ Position 7

14

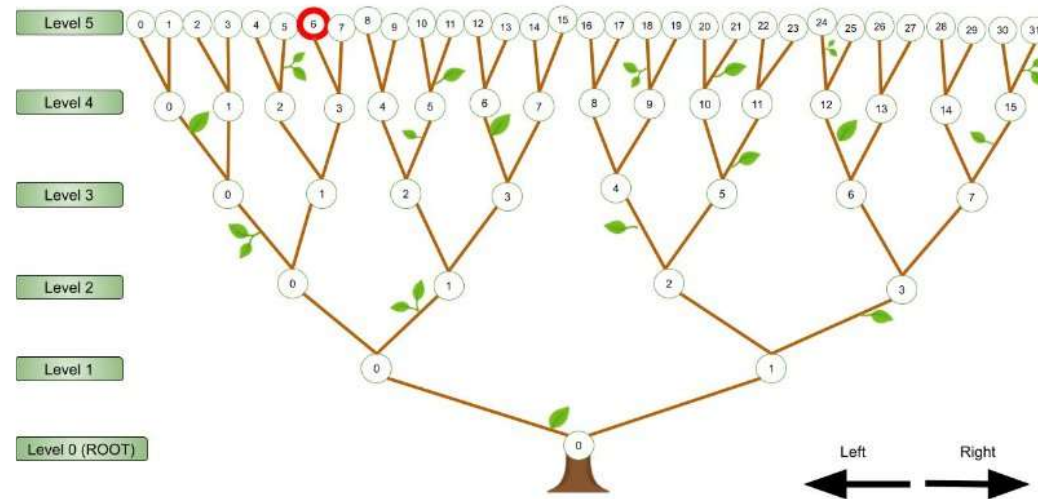
In the tree shown **BELLOW**, at which **LEVEL** is the agent currently located? *



- ☐ Level 1
- ☐ Level 2
- ☐ Level 3
- ☐ Level 4

15

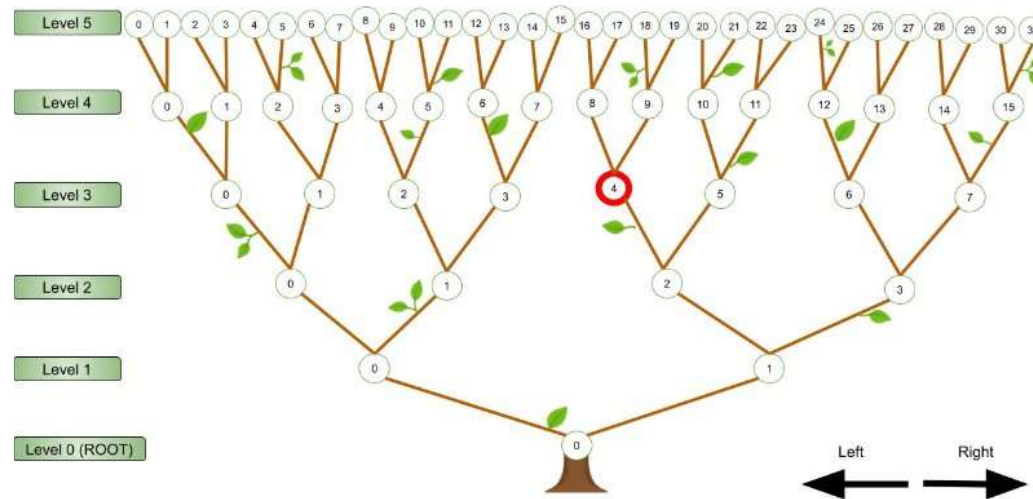
In the tree shown **BELLOW**, at which **LEVEL** is the agent currently located? *



- ☐ Level 2
- ☐ Level 3
- ☐ Level 4
- ☐ Level 5

16

In the tree shown **BELLOW**, at which **POSITION** is the agent currently located? *



- ☐ Position 0
- ☐ Position 3
- ☐ Position 4
- ☐ Position 7

Section 2

For this second part, you will be shown observations of the environment and the agent, together with text sentences describing the overall behavior of one agent with a fixed set of preferences for nutritional values. You will be asked to predict the agent's behavior based on these descriptions.

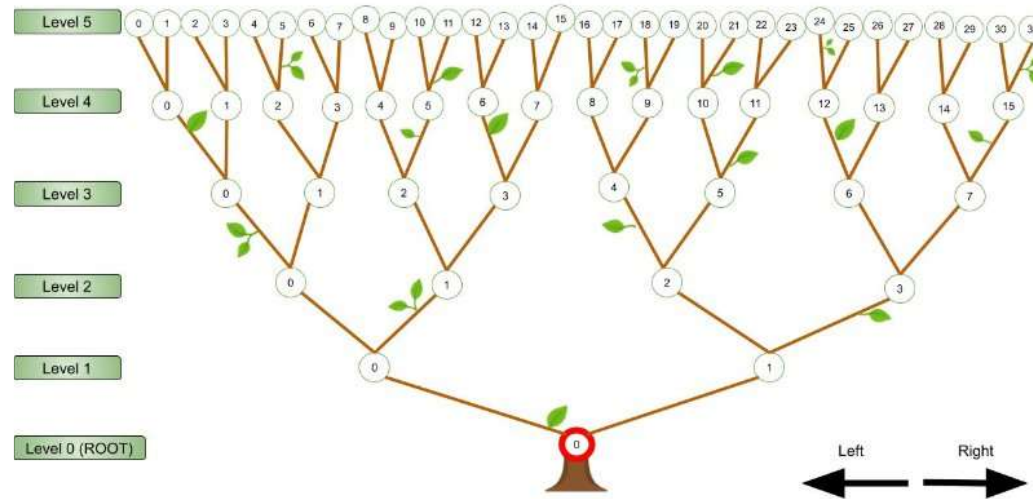
The state conditions in these descriptions can take two forms:

- **Exact value:** for example, "LEVEL is equal to 4," which applies only when $LEVEL = 4$.
- **Range of values:** for example, "LEVEL is between 0 and 2," which **includes both endpoints**, meaning that LEVEL can be 0, 1 or 2.

17

"When **LEVEL** is between 0 and 1, the agent has a **0.9** probability of going **LEFT**."

Which is the most probable action the agent will take in this situation? *

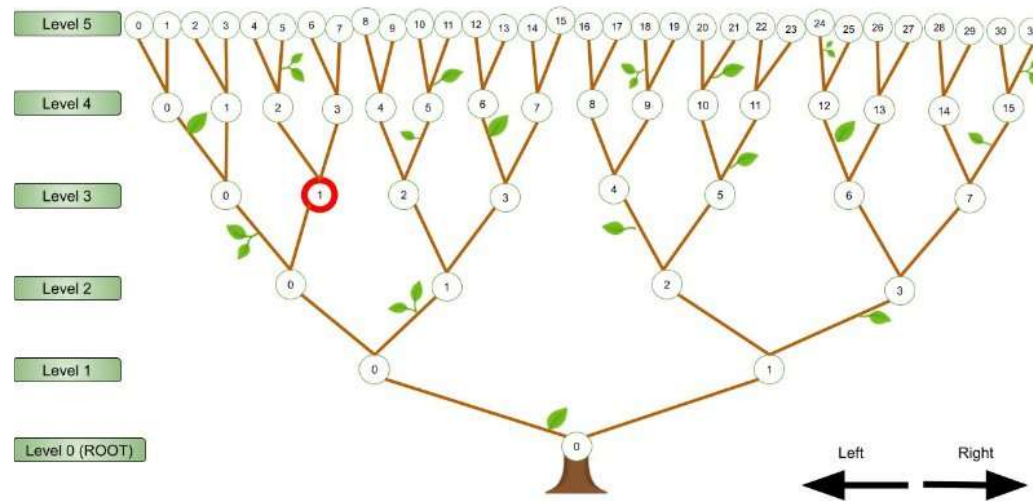


- ☐ Going Left.
- ☐ Going Right.

18

"When **LEVEL** is equal to 3, the agent has a **0.4** probability of going **RIGHT**."

Which is the most probable action the agent will take in this situation? *

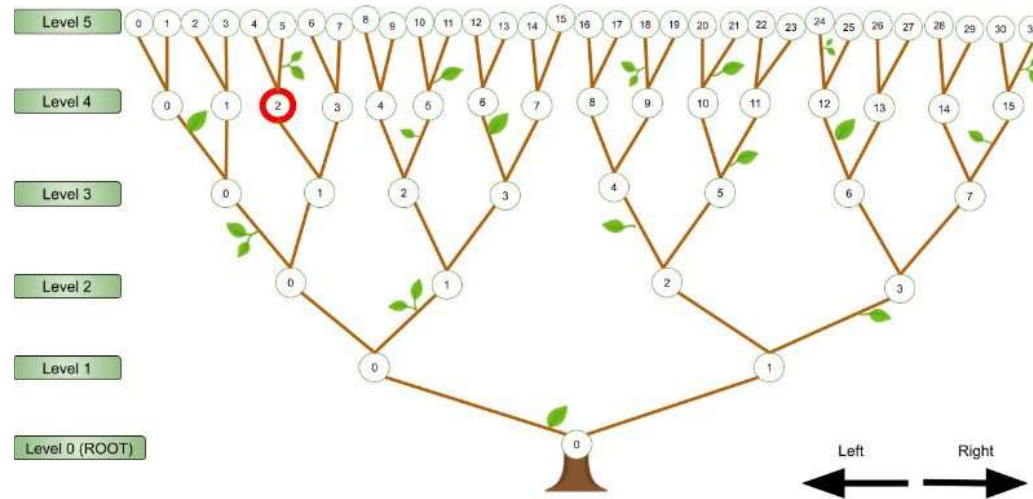


- ☐ Going Left.
- ☐ Going Right.

19

"When **LEVEL** is between 3 and 4, the agent has a **0.9** probability of going **LEFT**."

Which is the most probable action the agent will take in this situation? *

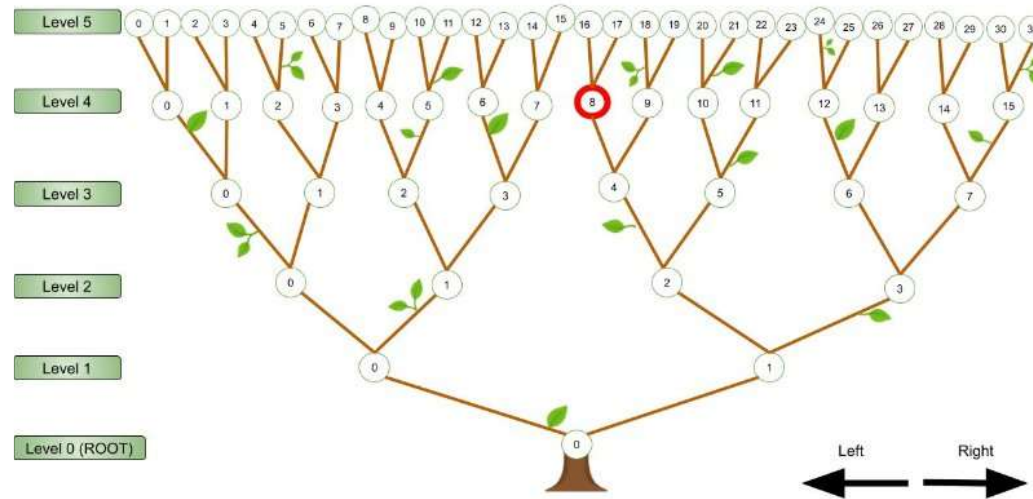


- ☐ Going Left.
- ☐ Going Right.

20

"When **POSITION** is between 8 and 12, the agent has a **0.3** probability of going **RIGHT**."

Which is the most probable action the agent will take in this situation? *

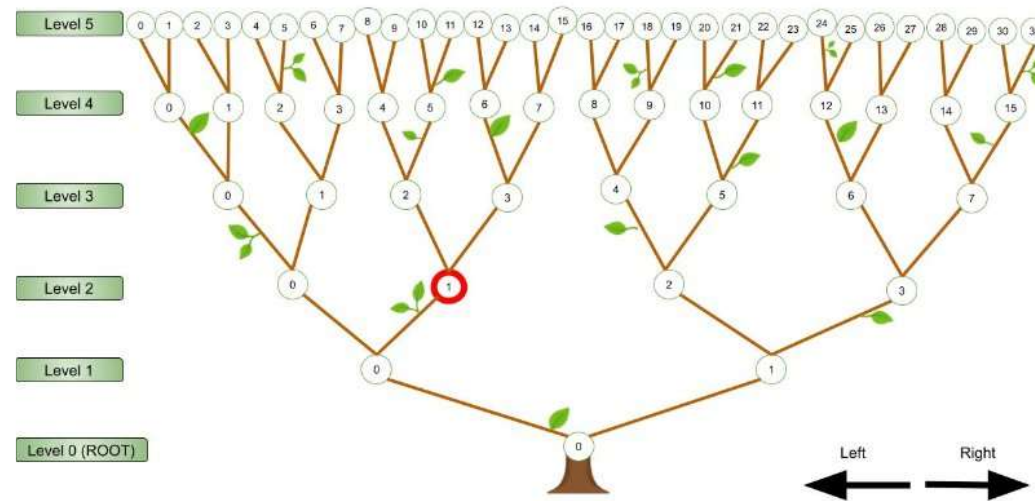


- ☐ Going Left.
- ☐ Going Right.

21

"When **POSITION** is equal to 1, the agent has a **1.0** probability of going **RIGHT**."

Which is the most probable action the agent will take in this situation? *

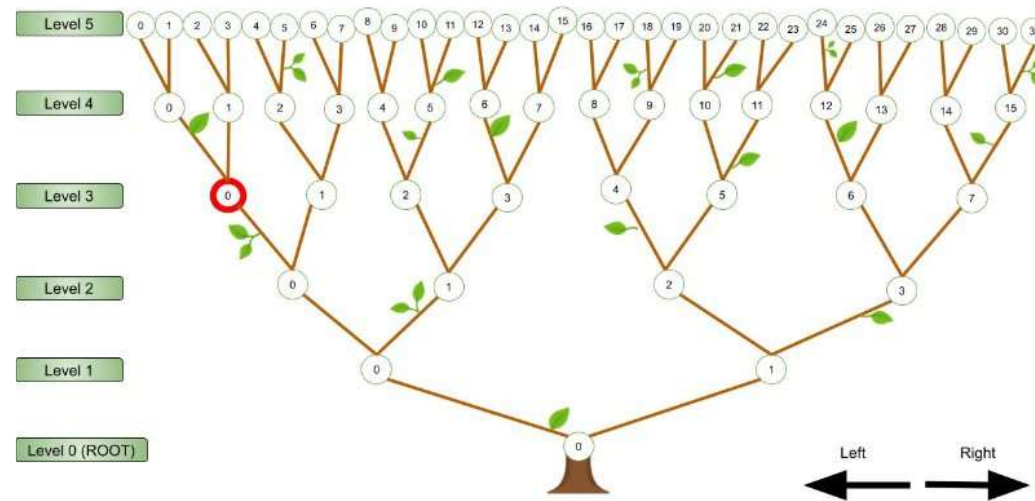


- ☐ Going Left.
- ☐ Going Right.

22

"When **POSITION** is between 0 and 1, the agent has a **0.9** probability of going **LEFT**."

Which is the most probable action the agent will take in this situation? *

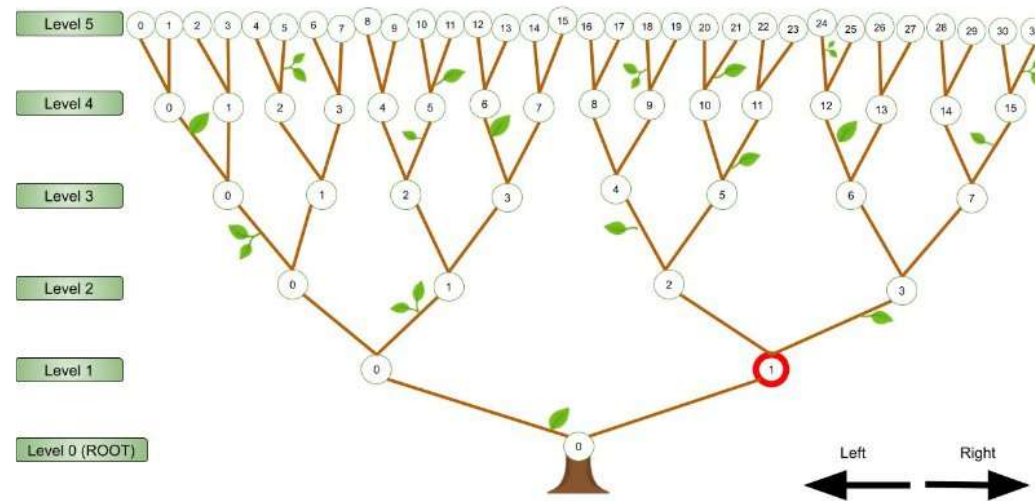


- ☐ Going Left.
- ☐ Going Right.

23

"When **LEVEL** is between 0 and 1, the agent has a **0.9** probability of going **RIGHT**."

Which is the most probable action the agent will take in this situation? *

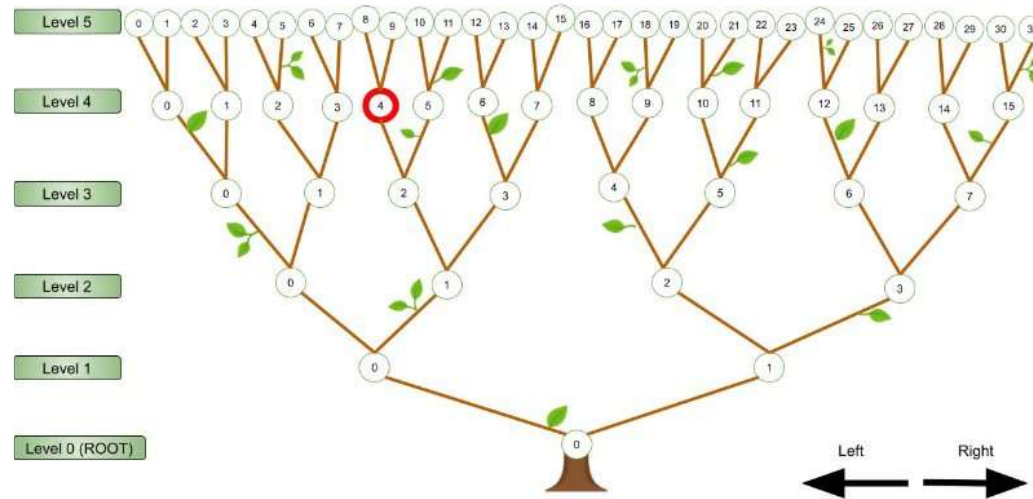


- ☐ Going Left.
- ☐ Going Right.

24

"When **LEVEL** is equal to 4, the agent has a **0.9** probability of going **left**."

Which is the most probable action the agent will take in this situation? *

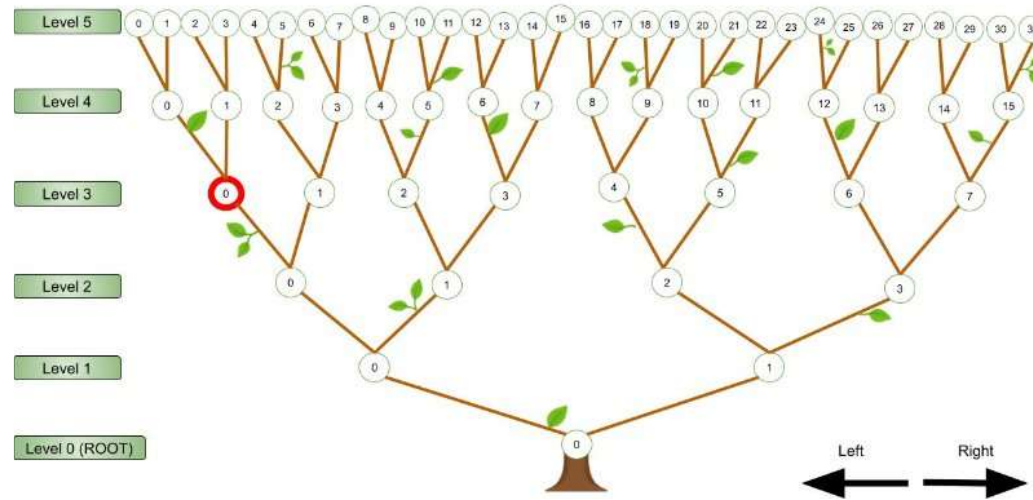


- ☐ Going Left.
- ☐ Going Right.

25

"When **LEVEL** is between 2 and 4, the agent has a **0.4** probability of going **RIGHT**."

Which is the most probable action the agent will take in this situation? *

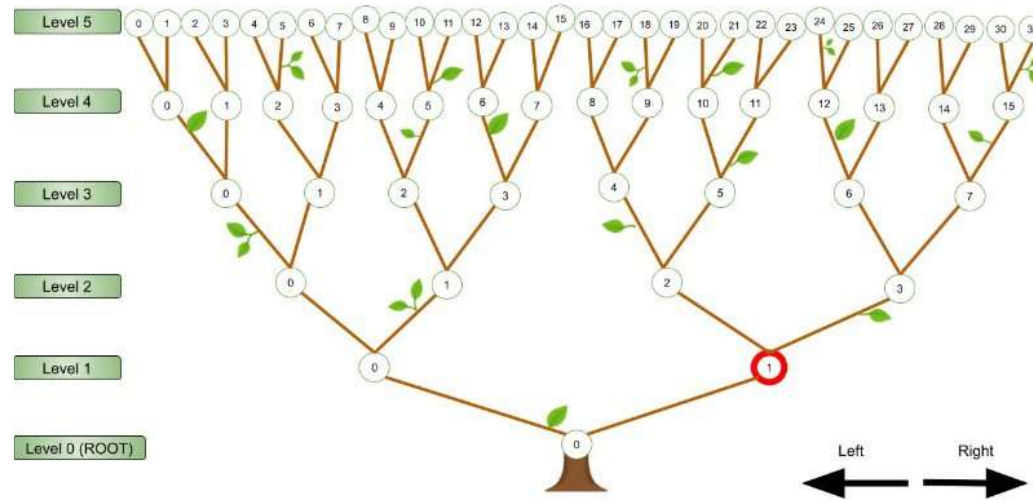


- ☐ Going Left.
- ☐ Going Right.

26

"When **LEVEL** is equal to 1, the agent has a **0.8** probability of going **LEFT**."

Which is the most probable action the agent will take in this situation? *



- ☐ Going Left.
- ☐ Going Right.

Section 3

For this third part, you will now be shown a contrastive explanation comparing two different AI agents' behaviour in the same environment with differing priorities. The two agents act in the same fruit-tree environment but have different preferences for the nutritional values of the fruits. You will be shown observations of the environment and the agents again, as in the previous section.

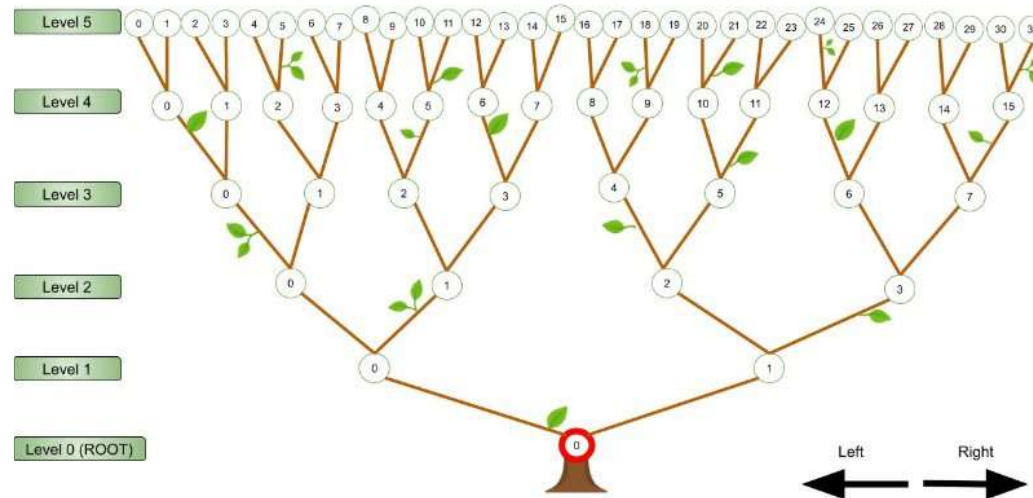
You will be asked to predict the behavior of both agents based on these explanations.

Note: For this study, some action probabilities may have been changed from the ones generated by the original algorithm

27

"When the agent's **POSITION** is between 0 and 1, **Agent A** has a **1.0** probability of going **RIGHT**, while **Agent B** has a **0.0** probability of going **RIGHT**."

Which is the most probable action **Agent B** will take in this situation? *

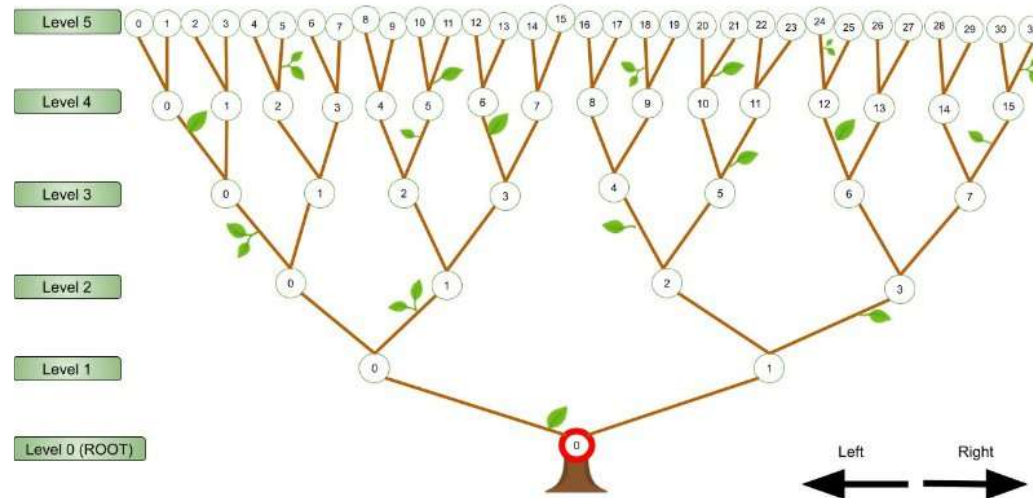


- ☐ Going Left.
- ☐ Going Right.

28

"When the agent's **POSITION** is between 0 and 1, **Agent A** has a **1.0** probability of going **RIGHT**, while **Agent B** has a **0.0** probability of going **RIGHT**."

Which is the most probable action **Agent A** will take in this situation? *

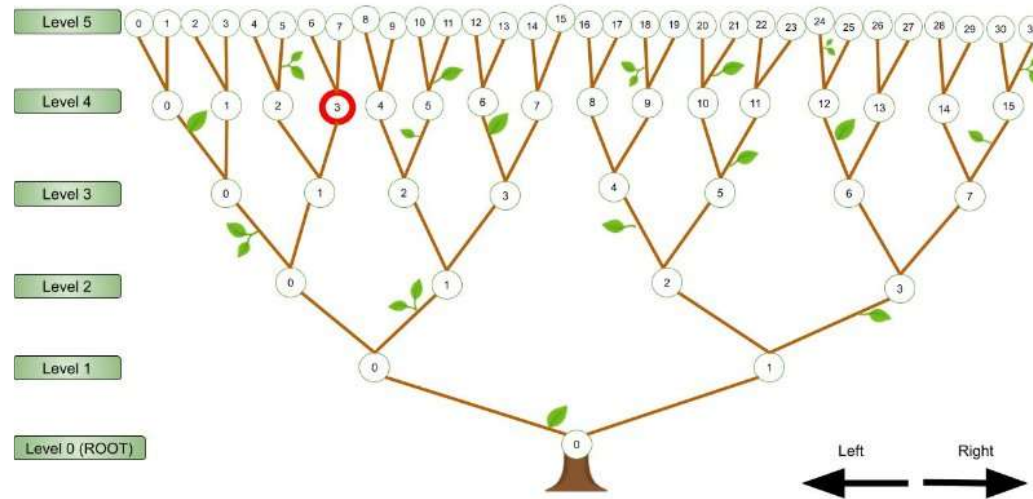


- ☐ Going Left.
- ☐ Going Right.

29

"When the agent's **LEVEL** is equal to **4**, **Agent A** has a **0.7** probability of going **RIGHT**, while **Agent B** has a **0.2** probability of going **RIGHT**."

Which is the most probable action **Agent A** will take in this situation? *

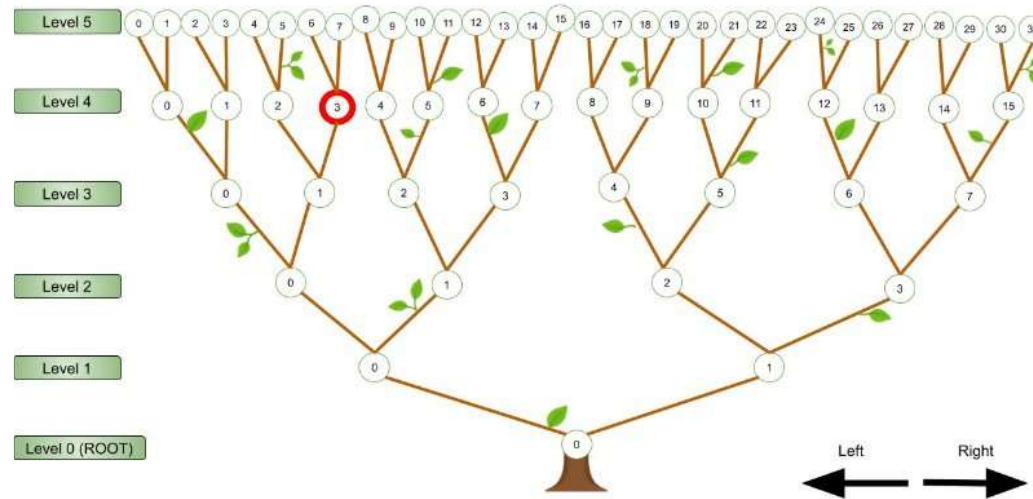


- ☐ Going Left.
- ☐ Going Right.

30

"When the agent's **LEVEL** is equal to **4**, **Agent A** has a **0.7** probability of going **RIGHT**, while **Agent B** has a **0.2** probability of going **RIGHT**."

Which is the most probable action **Agent A** will take in this situation? *

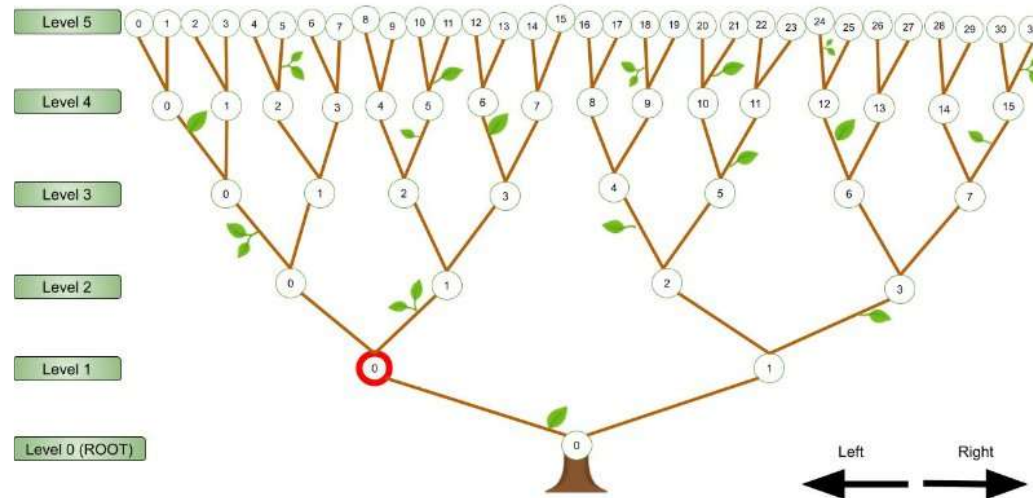


- ☐ Going Left.
- ☐ Going Right.

31

"When the agent's **POSITION** is equal to 1, **Agent A** has a **0.9** probability of going **RIGHT**, while **Agent B** has a **0.3** probability of going **RIGHT**."

Which is the most probable action **Agent A** will take in this situation? *

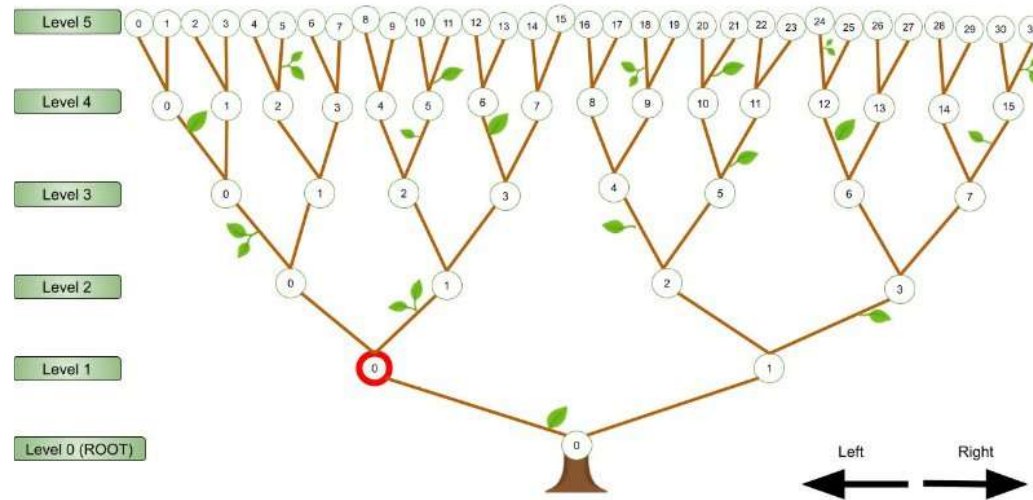


- ☐ Going Left.
- ☐ Going Right.

32

"When the agent's **POSITION** is equal to **1**, **Agent A** has a **0.9** probability of going **RIGHT**, while **Agent B** has a **0.3** probability of going **RIGHT**."

Which is the most probable action **Agent B** will take in this situation? *

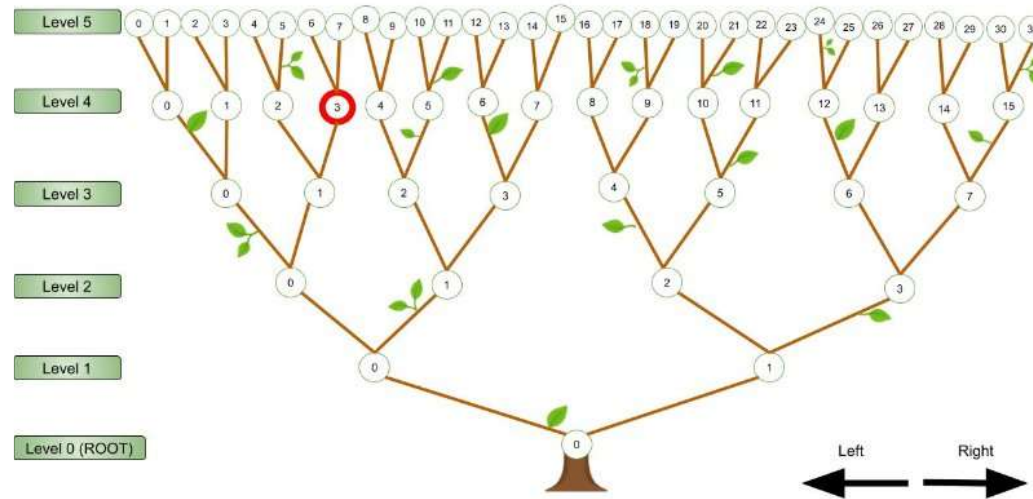


- ☐ Going Left.
- ☐ Going Right.

33

"When the agent's **LEVEL** is equal to **4**, **Agent A** has a **0.0** probability of going **LEFT**, while **Agent B** has a **1.0** probability of going **LEFT**."

Which is the most probable action **Agent A** will take in this situation? *

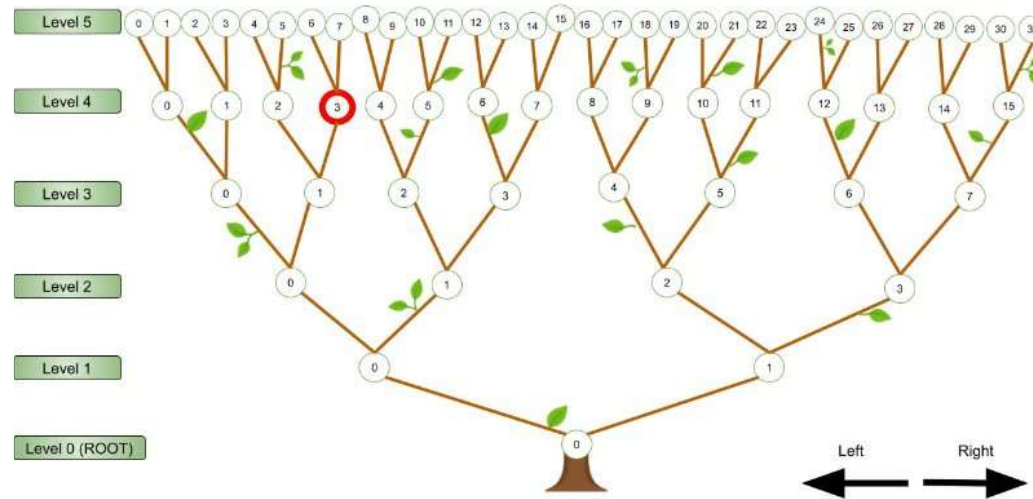


- ☐ Going Left.
- ☐ Going Right.

34

"When the agent's **LEVEL** is equal to **4**, **Agent A** has a **0.0** probability of going **LEFT**, while **Agent B** has a **1.0** probability of going **LEFT**."

Which is the most probable action **Agent B** will take in this situation? *

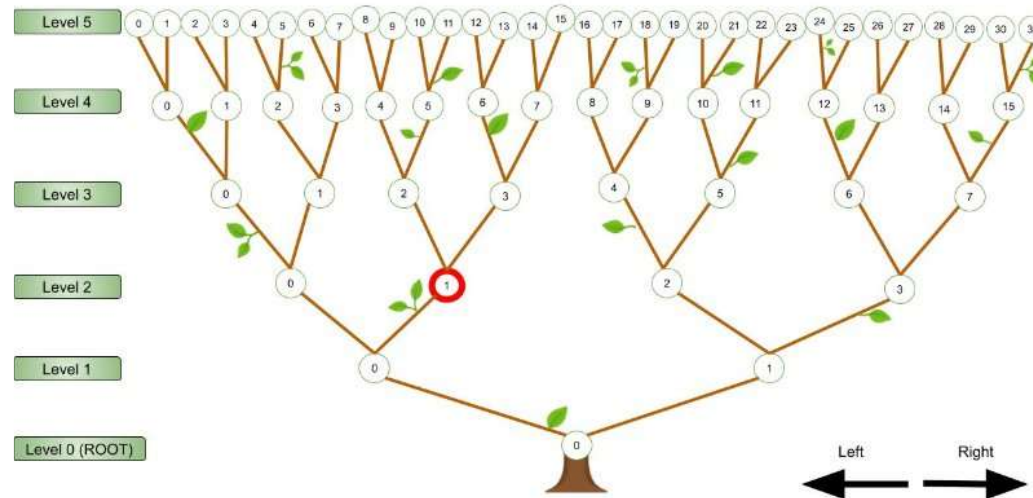


- ☐ Going Left.
- ☐ Going Right.

35

"When the agent's **LEVEL** is between **1** and **2**, **Agent A** has a **0.6** probability of going **RIGHT**, while **Agent B** has a **0.1** probability of going **RIGHT**."

Which is the most probable action **Agent A** will take in this situation? *

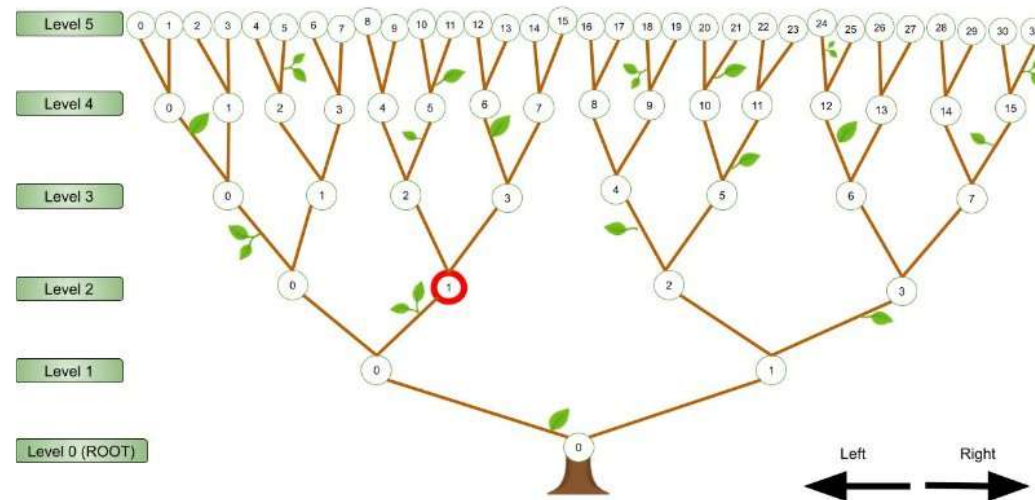


- ☐ Going Left.
- ☐ Going Right.

36

"When the agent's **LEVEL** is between **1** and **2**, **Agent A** has a **0.6** probability of going **RIGHT**, while **Agent B** has a **0.1** probability of going **RIGHT**."

Which is the most probable action **Agent B** will take in this situation? *

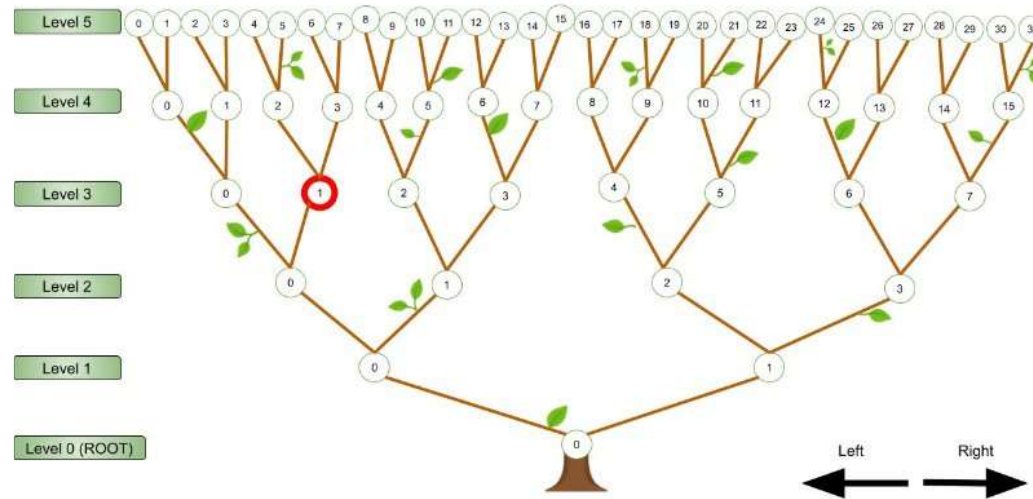


- ☐ Going Left.
- ☐ Going Right.

37

"When the agent's **LEVEL** is equal to **3**, **Agent A** has a **1.0** probability of going **RIGHT**, while **Agent B** has a **0.0** probability of going **RIGHT**."

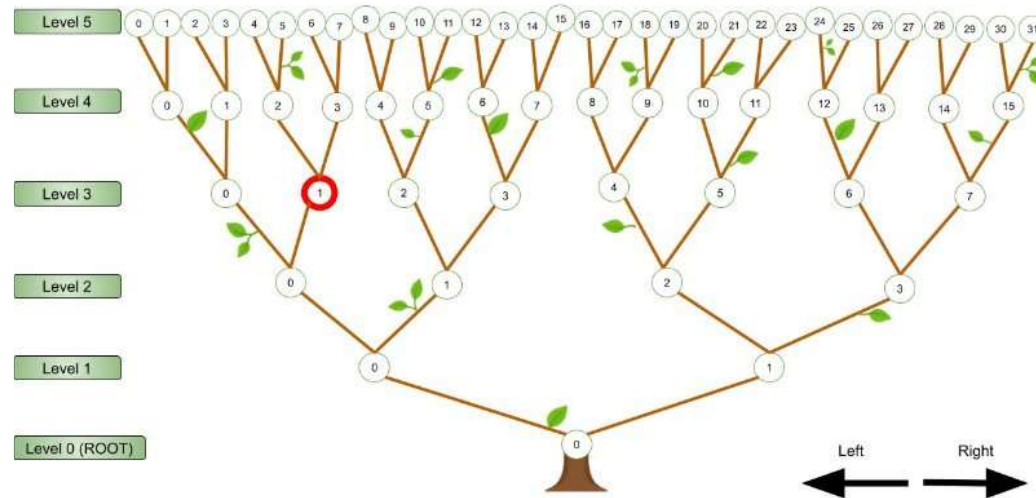
Which is the most probable action **Agent A** will take in this situation? *



- ☐ Going Left.
- ☐ Going Right.

38

"When the agent's **LEVEL** is equal to **3**, **Agent A** has a **1.0** probability of going **RIGHT**, while **Agent B** has a **0.0** probability of going **RIGHT**."
Which is the most probable action **Agent B** will take in this situation? *

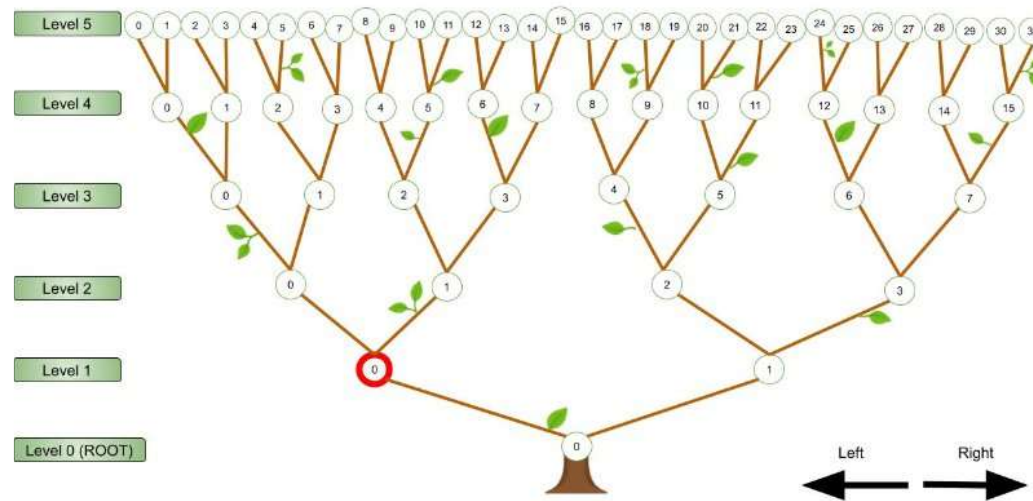


- ☐ Going Left.
- ☐ Going Right.

39

"When the agent's **LEVEL** is between **1** and **2**, **Agent A** has a **0.2** probability of going **LEFT**, while **Agent B** has a **0.9** probability of going **LEFT**."

Which is the most probable action **Agent A** will take in this situation? *

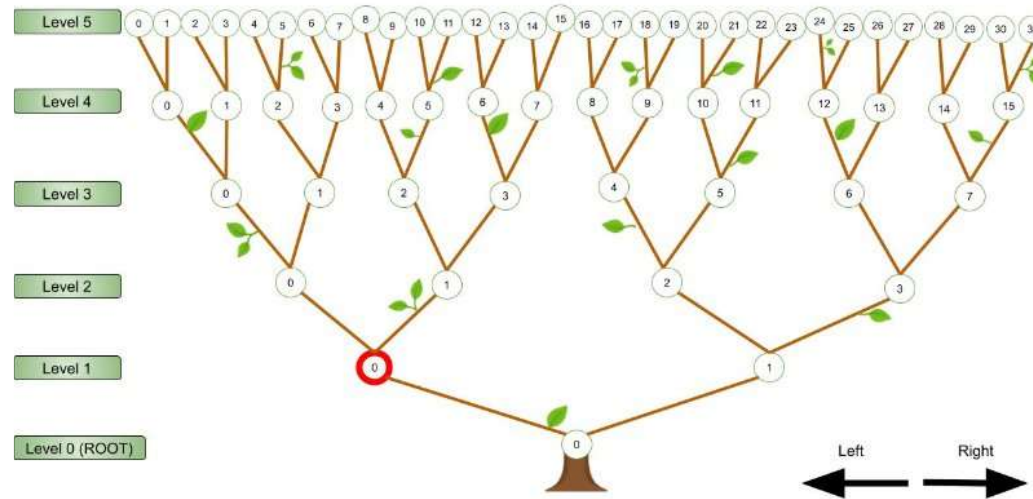


- ☐ Going Left.
- ☐ Going Right.

40

"When the agent's **LEVEL** is between **1** and **2**, **Agent A** has a **0.2** probability of going **LEFT**, while **Agent B** has a **0.9** probability of going **LEFT**."

Which is the most probable action **Agent B** will take in this situation? *

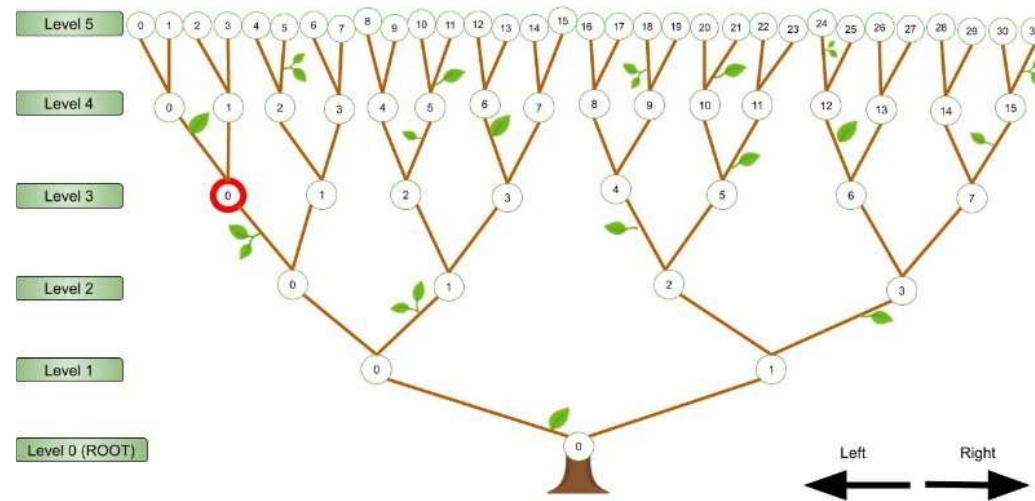


- ☐ Going Left.
- ☐ Going Right.

41

"When the agent's **LEVEL** is equal to **3**, **Agent A** has a **0.0** probability of going **LEFT**, while **Agent B** has a **1.0** probability of going **LEFT**."

Which is the most probable action **Agent B** will take in this situation? *

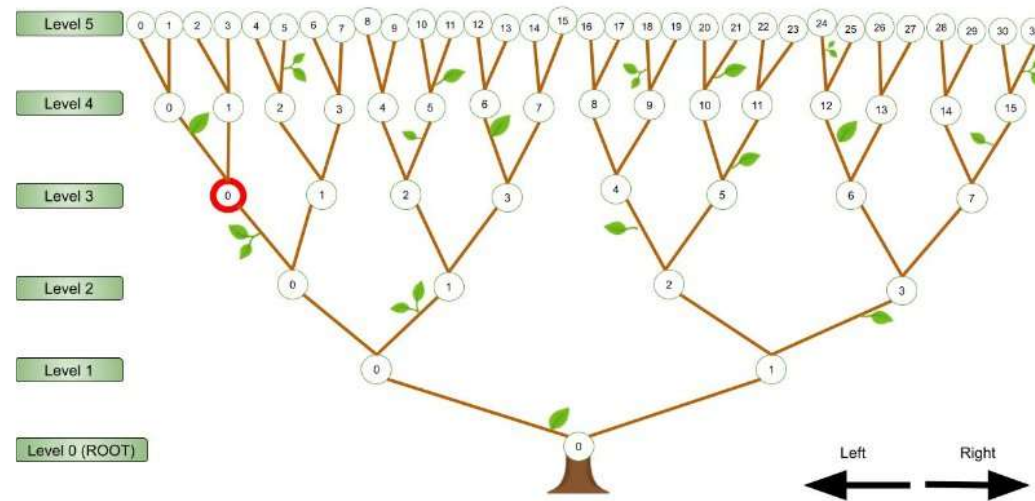


- ☐ Going Left.
- ☐ Going Right.

42

"When the agent's **LEVEL** is equal to **3**, **Agent A** has a **0.0** probability of going **LEFT**, while **Agent B** has a **1.0** probability of going **LEFT**."

Which is the most probable action **Agent A** will take in this situation? *

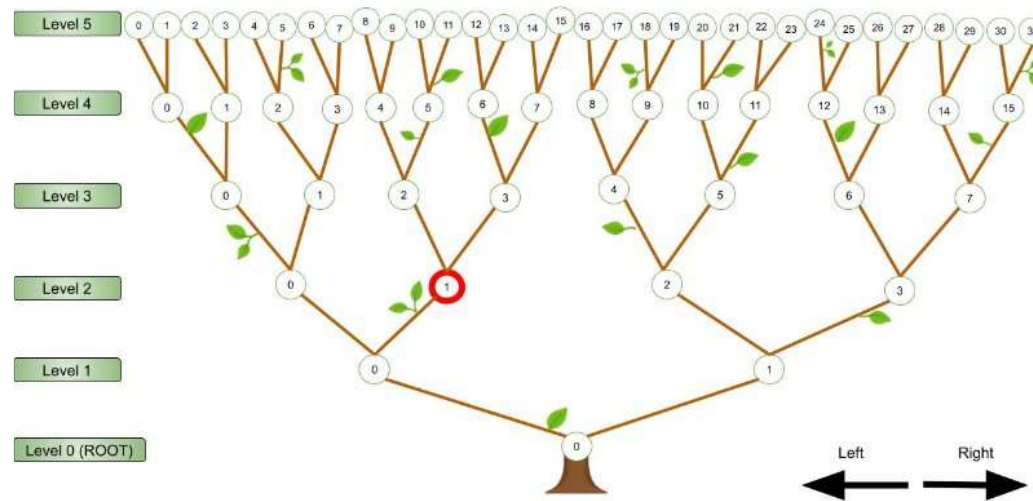


- ☐ Going Left.
- ☐ Going Right.

43

"When the agent's **LEVEL** is equal to **2**, **Agent A** has a **0.8** probability of going **RIGHT**, while **Agent B** has a **0.1** probability of going **RIGHT**."

Which is the most probable action **Agent B** will take in this situation? *

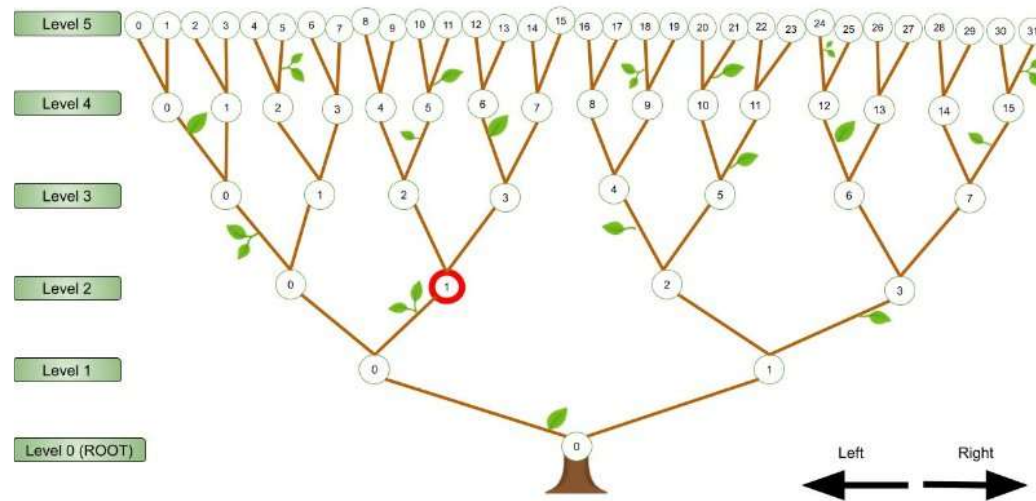


- ☐ Going Left.
- ☐ Going Right.

44

"When the agent's **LEVEL** is equal to **2**, **Agent A** has a **0.8** probability of going **RIGHT**, while **Agent B** has a **0.1** probability of going **RIGHT**."

Which is the most probable action **Agent A** will take in this situation? *



- ☐ Going Left.
- ☐ Going Right.

Section 4

In this part you will be asked to give an assessment of the usefulness, clarity and understandability of the explanations of **PART 3**, you will be given statements and will have to rate from 1 to 5 how much you are in agreement with them, 1 being I strongly disagree completely and 5 being I strongly agree.

45

"The explanation helped me understand the differences between the agents' behavior" *

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

46

"I find the explanations of how the agents operate satisfying" *

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

47

"The explanations are sufficiently detailed" *

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

48

"The explanations are sufficiently complete" *

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

49

"The explanations are actionable, that is, help me know how to use the agents" *

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

50

"The explanations let me know how reliable the agents are" *

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

51

"The explanations let me know how trustworthy the agents are" *

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree